

RESEARCH INTERESTS

I am passionate about advancing foundational research in machine learning, focusing on self-supervised learning and generative modeling. My work spans self-supervised representation learning, 3D object reconstruction and generation, and exploring the interplay between these areas. I am eager to apply my expertise across diverse research domains to drive innovation and address complex challenges.

EDUCATION

- **The University of Edinburgh** Edinburgh, UK
Doctor of Philosophy in Computer Science *Sep. 2021 – Mar 2025(Expected)*
- **Technical University of Munich** Munich, Germany
Master of Science in Computer Science *Oct. 2018 – Dec. 2020*
- **Istanbul Technical University** Istanbul, Turkey
Bachelor of Science in Computer Science *Sep. 2013 – June. 2017*

EXPERIENCE

- **Meta AI** California, USA
Research Scientist Intern *Aug 2023 – Feb 2024*
 - **3D and Generative Learning:** Developed methods to integrate self-supervised learning and 3D generative modeling, focusing on mutual benefits for reconstruction and synthesis tasks.
- **Institute for Adaptive and Neural Computation** Edinburgh, UK
Postgraduate Researcher *Sep 2021 – Present*
 - **Shape and Semantics in ML:** Developed models for articulated 3D object reconstruction and explored the role of shape in semantic tasks, including fine-grained classification and correspondence.
- **TUM Computer Vision Lab** Munich, Germany
Research Assistant *Oct 2018 – Jun 2021*
 - **Holistic Scene Understanding:** Worked on Lidar-based semantic segmentation, instance segmentation, and multi-object tracking.
 - **Unsupervised Shape Analysis:** Explored self-supervised approaches for shape correspondence using functional maps and heat kernels.

PUBLICATIONS

- D. Duolikun, **M. Aygun**, C. Li, H. Bilen, and O. Mac Aodha. "DepthCues: Evaluating Monocular Depth Perception in Large Vision Models." **arXiv:2411.17385 (2024)**
- **M. Aygun**, O. Mac Aodha, "SAOR: Single-view Articulated Object Reconstruction", **CVPR 24'**
- **M. Aygun**, P. Dhar, Z. Yan, O. Mac Aodha, and R. Ranjan, "Enhancing 2D Representation Learning with a 3D Prior ", **CVPR 24' - Workshop on Representation Learning with Very Limited Images'**
- **M. Aygun**, O. Mac Aodha, "Demystifying Unsupervised Semantic Correspondence Estimation", **ECCV 22'**
- **M. Aygun**, A. Osep, M. Weber, M. Maximov, C. Stachniss, J. Behley, L. Lael-Taixe, "4D Panoptic Lidar Segmentation", **CVPR 21'**
- **M. Aygun**, Z. Lahner, D. Cremers, "Unsupervised Dense Shape Correspondence using Heat Kernels", **3DV 20'**
- **M. Aygun**, Y. Aytar, H. K. Ekenel, "Exploiting Convolution Filter Patterns for Transfer Learning", **ICCV 17' Workshop on Transfer Learning - (Honourable Mention Award)**

SKILLS

- Technical: Python, PyTorch (3D), C/C++, OpenCV, NumPy, Matlab, Linux, Git
- Methodologies: Generative modeling, self-supervised learning, 3D reconstruction, shape analysis

AWARDS & ACTIVITIES

- Reviewer for TPAMI, CVPR, ICCV, ECCV
- Tutoring for Applied Machine Learning Course at the University of Edinburgh. 2021-2024
- DAAD Scholarships for Master Studies in Germany, 2018-2020

REFERENCES

- **Dr. Oisín Mac Aodha**, The University of Edinburgh (oisin.macaodha@ed.ac.uk)
- **Prof. Laura Leal-Taixe**, Nvidia (leal.taixe@tum.de)